

Eco-Loop Building Agents

Autonomous, MCP-driven closed-loop energy optimization for buildings

+6.4% ENERGY SAVINGS

Problem Statement: Eco-Loop Building Agents

Theme: Physical AI / Smart Building Automation

Category: Software

The Idea

A live, physics-grounded closed loop: EnergyPlus simulates a real building; an MCP tool server exposes its state and controls; an AI control layer reads sensor-like output, reasons against explicit comfort constraints, and writes real setpoint changes back — with no human editing code between iterations.

Closed-loop, not scripted

The building is edited and re-simulated automatically based on live tool-read results — not a fixed rule schedule.

Constraint-first design

Thermal comfort (PMV -0.5 to +0.5) is a hard constraint the optimizer cannot trade away, not a soft preference.

Honest, dual-path autonomy

An LLM agent demonstrates real MCP tool-calling; a deterministic search guarantees the reported savings are reproducible.

Technical Approach

Stack

- EnergyPlus 26.1.0 — physics-based annual simulation, Mumbai ISHRAE weather
- eppy — Python/IDF bridge for reading & editing the building model
- MCP-style tool server (FastMCP) — `get_current_summary`, `set_zone_setpoint`, `run_simulation`
- Groq API, llama-3.1-8b-instant — open-source LLM control agent
- Deterministic sweep controller — guaranteed, reproducible savings

Closed-Loop Flow

1. Read: `get_current_summary()` — monthly kWh + PMV comfort
2. Reason: evaluate vs. hard comfort constraint + energy objective
3. Act: `set_zone_setpoint()` — safety-bounded, setback-preserving edit
4. Verify: `run_simulation()` — full annual re-run, real result
5. Repeat until constraint-satisfying optimum found

Key engineering safeguards

Server-enforced safety bounds (heating 16–24°C, cooling 22–30°C, 2°C min deadband) — not left to the LLM.

Setback-preserving edits — only occupied-hours setpoints change; night/weekend setback logic is untouched.

Rate-limit resilient agent loop — backoff + retry so a transient 429 never discards a real applied change.

Feasibility & Viability

Proven Feasible — Real Results

- Full annual EnergyPlus simulation completes in ~9-16s per iteration — fast enough for iterative search
- MCP-style tool interface works end-to-end with both an LLM agent and a deterministic controller
- 78,057.8 kWh baseline → 73,040.0 kWh optimized: a real, reproducible +6.4% saving, independently re-run and confirmed

Known Challenges & Mitigations

- 8B open-source LLM doesn't reliably follow multi-step directional feedback → deterministic sweep guarantees a defensible result regardless
- Free-tier LLM API rate limits (6000 TPM) → inter-call backoff + single retry added
- Energy savings trade off against comfort → explicit 20%-of-baseline comfort budget makes the trade-off transparent, not hidden

78,057.8

Baseline kWh/yr

73,040.0

Optimized kWh/yr

+6.4%

Energy Savings

38 → 45

Comfort Violations

Artifacts

Deliverables produced

Source Code

mcp_energyplus_server.py, run_agent_loop.py, run_hillclimb_loop.py, run_mumbai_baseline.py, summarize_baseline.py

Building Models

Baseline (frozen) + AI-optimized .idf, plus the full 5-point setpoint sweep

Quantitative Dashboard

Interactive HTML dashboard: trade-off chart, full results table, methodology

Architecture Document

Markdown report: tool design, prompt engineering iterations, latency handling, log summarization

Research & References

- DOE EnergyPlus documentation — simulation engine & IDF schema reference
- eppy documentation — Python bridge for EnergyPlus IDF manipulation
- ASHRAE 55 — Predicted Mean Vote (PMV) thermal comfort model & acceptable band
- Model Context Protocol (MCP) specification — standardized tool-calling interface
- Groq API documentation — llama-3.1-8b-instant function-calling & rate limits
- ISHRAE Mumbai weather file (IND_Mumbai.430030_ISHRAE.epw) — local climate data